Marmara Üniversitesi
Electrical and Electronics Engineering Department
EE4092: Optoelectronics (Fall 2023)

Instructor Information	Mohammad A. Alsunaidi	Office M4-232	Phone ---	Office Hours Mon-Thu: 11:30-12:30	Email mohammad.alsunaidi@marmara	
Course Information	Text *Optoelectronics*, S. O. Kasap	Grading	Class Work Assignments + Exp (4x5)% + Project 10%		Midterm Exam 30%	Final Exam 40%

Week	Events	Subjects	Tuesdays	Wednesdays
1 Oct 2-6	---	Nature of Light and Material Properties	Course introduction and scope.	Light as an EM Wave: Maxwell's and Wave Equations. Types of waves.
2 Oct 9-13	---		Light Propagation equation.	Light Propagation in different media.
3 Oct 16-20	Course drop/add Week		Light Reflection and Refraction.	Light Reflection and Refraction. Anti-reflection Surfaces.
4 Oct 23-27	Cumhuriyet Bayramı Oct 29		Light-matter interaction	Light as an energy quanta: The photoelectric effect and the Photon.
5 Oct 30-Nov 3	---	The PN Junction	Semiconductor Fundamentals and the PN Junction. PN junction energy bands.	Photon absorption and generation. Direct and Indirect semiconductors.
6 Nov 6-10	Assignment 1 Due Nov 10	Photodetectors	Photodetectors types and basic structures.	Characteristics and modes of operation. PN, PiN, APD and Schottky PD.
7 Nov 13-17	---		Photodiode circuit configurations and modes of operation.	Phototransistor circuit configurations and modes of operation.
Nov 20-26	**Mid-term Exams period**			
9 Nov 27-Dec 1	Experiment 1 Due Nov 17	Solar Cells	Basic and improved solar cells structures and materials. Thin film cells. Design rules.	Photo-voltaic (PV) circuit analysis.
10 Dec 4-8	---	Light Emitting Diodes (LEDs)	LEDs basic design and operation.	LED characterization: optical power, spectral distribution and modulation BW.
11 Dec 11-15	Assignment 2 Due Dec 15		LED characterization: optical power, spectral distribution and modulation BW.	Improved LED structures and materials.
12 Dec 18-22	Midterm Make-up (Excused, Dec 21-23)		UV LEDs. Material systems, basic designs and challenges.	White LEDs. Techniques, efficiencies and challenges.
13 Dec 25-29	Experiment 2 Due Dec 29		LED circuits. Basic amplifier configurations.	LED circuits. Modules, and panels.
14 Jan 1-5	New Year 2024 Monday	Opto-couplers and Opto-isolators	Opto-couplers basic design and applications.	Opto-couplers circuits.
15 Jan 8-12	Project submission Jan 12	Special topics	Students' Projects	Students' Projects
Jan 15-28	**Final Exams period**		**Jan 29-Feb 4 (Re-sit Final Exams period)**	